



L19: Events

Interactive Security Grid

Neon City Cyber Academy

*Module 5: Object-Oriented
Programming*





What are Events?

Events = Things that happen (clicks, keypresses, timer ticks)

Event-Driven Programming = Code that WAITS for events, then RESPONDS

Examples:

-  Click a button → Action happens
-  Press a key → Character moves
-  Timer expires → Game over



The Three Event Methods

```
screen = turtle.Screen()

# 1. Listen for keyboard events
screen.listen()

# 2. Bind a key to a function
screen.onkey(function_name, 'key')

# 3. Bind mouse click to a function
screen.onclick(function_name)
```

Critical: Call `screen.listen()` FIRST!



Task 1: Arrow Key Movement

```
import turtle

guard = turtle.Turtle()
guard.shape('turtle')

def move_forward():
    guard.forward(20)

screen = turtle.Screen()
screen.listen()
screen.onkey(move_forward, 'Up')

turtle.mainloop()
```



Key Concepts

Function vs Function Call:

```
# ❌ WRONG - Calls immediately  
screen.onkey(move_forward(), 'Up')
```

```
# ✅ CORRECT - Passes function reference  
screen.onkey(move_forward, 'Up')
```

"No parentheses! You're giving Python the RECIPE, not cooking it immediately!"



Mouse Click Events

```
stealth_drone = turtle.Turtle()

def change_color(x, y):
    stealth_drone.color('lime')
    stealth_drone.goto(x, y)

screen = turtle.Screen()
screen.onclick(change_color)

turtle.mainloop()
```

Click anywhere → Drone teleports AND changes color!



Understanding (x, y) Parameters

onclick functions automatically receive coordinates:

```
def my_function(x, y):  
    print(f"Clicked at: ({x}, {y})")
```

- **x** = Horizontal position of click
- **y** = Vertical position of click

You don't pass them - Python does automatically!



4-Direction Control (WASD)

```
patrol_unit = turtle.Turtle()
```

```
def up():  
    patrol_unit.forward(30)
```

```
def down():  
    patrol_unit.backward(30)
```

```
def left():  
    patrol_unit.left(45)
```

```
def right():  
    patrol_unit.right(45)
```

```
screen = turtle.Screen()  
screen.listen()  
screen.onkey(up, 'w')  
screen.onkey(down, 's')  
screen.onkey(left, 'a')  
screen.onkey(right, 'd')
```




Event Loop Explained

```
turtle.mainloop()
```

What this does:

1. Keeps program running
2. Constantly checks for events
3. Calls your functions when events occur
4. Repeats forever (until you close window)

Without `mainloop()`, program exits immediately!

✓ Lesson Complete!

You Mastered:

- ✓ Event-driven programming concept
- ✓ `screen.listen()` activation
- ✓ `onkey()` for keyboard events
- ✓ `onclick()` for mouse events
- ✓ Function references vs function calls
- ✓ Building interactive programs

Next: L20 - Paint Project



Challenge

Build a Security Keypad:

- 4 arrow keys for movement
- Space bar to toggle pen up/down
- 'c' key to clear screen
- Mouse click to change color

Make it fully interactive!